

Nandakumar Sasidharan Rajalekshmi

Researcher — In-Memory Computing & AI Hardware

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Date of birth 29 April 1991
Nationality Indian | Work permit: Swiss B (valid)



Professional Profile

Electrical engineer and researcher with over seven years of expertise in **in-memory computing**, **neuromorphic engineering**, and **AI hardware acceleration**. Proven ability to bridge device physics and large-scale neural network deployment – from nanoscale PCM device fabrication to the first demonstration of floating-point-comparable on-chip training with 0.4 million analog memory devices. Holder of 3 granted US patents; co-author of 25+ peer-reviewed publications including *Nature Communications*, *IEEE Micro*, and *Scientific Reports*. Actively seeking industry R&D roles in Switzerland.

Work Experience

Postdoctoral Researcher IBM Research – Zurich, Switzerland Apr 2019 - Mar 2021

- Led experimental validation of mixed-precision training on 0.4 million PCM devices – first demonstration of floating-point-comparable on-chip training from analog hardware.
- Investigated in-memory computing for homomorphic encryption acceleration, evaluating the possibility to accelerate privacy-preserving applications.
- Provided accurate PCM device models for IBM's Analog AI hardware challenge; evaluated image recognition and speech detection networks.
- Major contribution to the Analog AI Chip Cloud Demo – recognised with IBM's *Outstanding Technical Achievement Award* and *Research Division Award* (2019).

Research Intern IBM Research – Zurich, Switzerland May 2017 - Jul 2018

- Developed mixed-precision deep learning architectures combining high-precision digital accumulators with low-precision analog crossbar arrays; validated on convolutional neural networks.
- Designed adaptive scaling methods for crossbar memory arrays to maintain learning accuracy under hardware non-idealities.
- Demonstrated first single-PCM-synapse training at floating-point accuracy; built statistically accurate PCM programming models capturing drift and noise.

Research Intern IBM Research – Zurich, Switzerland Jul 2016 - Aug 2016

- Characterized large PCM array, made statistical models of the memory devices, evaluated their performance in artificial and brain-inspired spiking neural networks.

Career Break – Independent Research & Teaching Apr 2021 - Present

- Completed 1750+hour intensive Classical Hatha Yoga Teacher Training (certified); organised workshops and retreats in India and Switzerland. Developed skills in programme design, independent teaching, and cross-cultural communication.
- Pursued independent research into the science of consciousness and contemplative traditions; produced an draft manuscript (available on request).
- Experience as breakfast cook and in house-keeping at a Seminar Hotel, Kientalerhof, Kiental.

Education

Ph.D. in Electrical Engineering New Jersey Institute of Technology (NJIT), NJ, USA GPA: 2016 - 2019
4.0 / 4.0

Dissertation: “High-performance learning systems using low-precision nanoscale devices”

Advisor: Prof. Bipin Rajendran (Northeastern University London, UK)

M.Tech. in Microelectronics and VLSI IIT Bombay, India CPI: 9.64 / 10 2013 - 2015

Technical Skills

Programming	Python, MATLAB, Verilog-A, HTML, CSS
AI / ML	Deep learning (training & inference), spiking neural networks, analog AI hardware co-design, statistical modelling
Simulation	COMSOL, NgSpice, Sentaurus, Cadence Spectre
Device Fab.	Experience with clean room device fabrication, Spin coating, e-beam lithography, sputter deposition, double-sided aligner, UV exposure, annealing
Characterisation	cAFM, B1500 IV system, WGF MU

Patents (US Granted)

1. **N. S. Rajalekshmi**, F. A. Bergamaschi, E. S. Eleftheriou. *Computation in Homomorphic Encryption Systems*. US Patent App. 17/321,538.
2. I. Boybat Kara, M. Le Gallo-Bourdeau, **N. S. Rajalekshmi**, A. Sebastian, E. S. Eleftheriou. *Mixed-Precision Deep Learning with Multi-Memristive Devices*. US Patent 11,188,825.
3. M. Le Gallo-Bourdeau, **N. S. Rajalekshmi**, C. Piveteau, I. Boybat Kara, A. Sebastian. *Training of Artificial Neural Networks*. US Patent 11,386,319 B2.

Peer-Reviewed Publications (25+ total; selected below)

Journals

1. Malte J. Rasch, Charles Mackin, Manuel Le Gallo, An Chen, Andrea Fasoli, Frédéric Odermatt, Ning Li, **S. R. Nandakumar**, Pritish Narayanan, Hsinyu Tsai, Geoffrey W. Burr, Abu Sebastian, and Vijay Narayanan. Hardware-aware training for large-scale and diverse deep learning inference workloads using in-memory computing-based accelerators. *Nature Communications*, 14:5282, 2023
2. Charles Mackin, Malte J. Rasch, An Chen, Jonathan Timcheck, Robert L. Bruce, Ning Li, Pritish Narayanan, Stefano Ambrogio, Manuel Le Gallo, **S. R. Nandakumar**, Andrea Fasoli, Jose Luquin, Alexander Friz, Abu Sebastian, Hsinyu Tsai, and Geoffrey W. Burr. Optimised weight programming for analogue memory-based deep neural networks. *Nature Communications*, 13:3765, 2022
3. Manuel Le Gallo, **S. R. Nandakumar**, Lazar Ciric, Irem Boybat, Riduan Khaddam-Aljameh, Charles Mackin, and Abu Sebastian. Precision of bit slicing with in-memory computing based on analog phase-change memory crossbars. *Neuromorphic Computing and Engineering*, 2022
4. **S. R. Nandakumar**, Manuel Le Gallo, Christophe Piveteau, Vinay Joshi, Giovanni Mariani, Irem Boybat, Geethan Karunaratne, Riduan Khaddam-Aljameh, Urs Egger, Anastasios Petropoulos, Theodore Antonakopoulos, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. Mixed-precision deep learning based on computational memory. *Frontiers in Neuroscience*, 14:406, 2020
5. Vinay Joshi, Manuel Le Gallo, Simon Haefeli, Irem Boybat, **S. R. Nandakumar**, Christophe Piveteau, Martino Dazzi, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. Accurate deep neural network inference using computational phase-change memory. *Nature Communications*, 11(1):2473, May 2020
6. **S. R. Nandakumar**, Irem Boybat, Manuel Le Gallo, Evangelos Eleftheriou, Abu Sebastian, and Bipin Rajendran. Experimental demonstration of supervised learning in spiking neural networks with phase-change memory synapses. *Scientific Reports*, 10(1):8080, May 2020
7. **S. R. Nandakumar** and Bipin Rajendran. Bio-mimetic synaptic plasticity and learning in a sub-500mV Cu/SiO₂/W memristor. *Microelectronic Engineering*, 226:111290, Apr 2020
8. **S. R. Nandakumar**, Manuel Le Gallo, Irem Boybat, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. A phase-change memory model for neuromorphic computing. *Journal of Applied Physics*, 124(15):152135, Oct 2018
9. **S. R. Nandakumar**, Shruti R. Kulkarni, Anakha V. Babu, and Bipin Rajendran. Building Brain-Inspired Computing Systems: Examining the Role of Nanoscale Devices. *IEEE Nanotechnology Magazine*, 12(3):19–35, Sep 2018
10. Irem Boybat, Manuel Le Gallo, **S. R. Nandakumar**, Timoleon Moraitis, Thomas Parnell, Tomas Tuma, Bipin Rajendran, Yusuf Leblebici, Abu Sebastian, and Evangelos Eleftheriou. Neuromorphic computing with multi-memristive synapses. *Nature Communications*, 9(1):2514, Jun 2018

11. **S. R. Nandakumar**, Marie Minvielle, Saurabh Nagar, Catherine Dubourdieu, and Bipin Rajendran. A 250 mV Cu/SiO₂/W Memristor with Half-Integer Quantum Conductance States. *Nano Letters*, 16(3):1602–1608, Mar 2016

Selected Conference Papers

1. **S. R. Nandakumar**, Irem Boybat, Jin-Ping Han, Stefano Ambrogio, Praneet Adusumilli, Robert L. Bruce, Matthew BrightSky, Malte Rasch, Manuel Le Gallo, and Abu Sebastian. Precision of synaptic weights programmed in phase-change memory devices for deep learning inference. In *2020 IEEE International Electron Devices Meeting (IEDM)*, pages 29.4.1–29.4.4, 2020
2. A. Sebastian, I. Boybat, M. Dazzi, I. Giannopoulos, V. Jonnalagadda, V. Joshi, G. Karunaratne, B. Kersting, R. Khaddam-Aljameh, **S. R. Nandakumar**, A. Petropoulos, C. Piveteau, T. Antonakopoulos, B. Rajendran, M. Le Gallo, and E. Eleftheriou. Computational memory-based inference and training of deep neural networks. In *2019 Symposium on VLSI Technology*, pages T168–T169, 2019
3. **S. R. Nandakumar**, Manuel Le Gallo, Irem Boybat, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. Mixed-precision architecture based on computational memory for training deep neural networks. In *International Symposium on Circuits and Systems (ISCAS)*, pages 1–5. IEEE, 2018
4. Michele Martemucci, Benedikt Kersting, Riduan Khaddam-Aljameh, Irem Boybat, **S. R. Nandakumar**, Urs Egger, Matthew Brightsky, Robert L Bruce, Manuel Le Gallo, and Abu Sebastian. Accurate weight mapping in a multi-memristive synaptic unit. In *2021 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 1–5. IEEE, 2021
5. **S. R. Nandakumar**, I. Boybat, M. Le Gallo, A. Sebastian, B. Rajendran, and E. Eleftheriou. Supervised learning in spiking neural networks with MLC PCM synapses. In *2017 75th Annual Device Research Conference (DRC)*, volume 22, pages 1–2. IEEE, Jun 2017
6. **S. R. Nandakumar** and Bipin Rajendran. Physics-based switching model for Cu/SiO₂/W quantum memristor. In *2016 74th Annual Device Research Conference (DRC)*, volume 16, pages 1–2. IEEE, Jun 2016

Awards & Honours

Outstanding Technical Achievement Award – Analog AI Chip Cloud Demo, IBM	2019
Research Division Award – Analog AI Chip Cloud Demo, IBM	2019
Hashimoto Prize for Doctoral Candidate in ECE, NJIT	2019
Outstanding Graduate Student Award, Newark College of Engineering, NJIT	2019
IBM Ph.D. Fellowship	2017-18
All India Rank 12 – GATE (Electronics & Communication), 250,000 candidates	2013
99.75 percentile – Kerala Engineering Entrance Examination	2009

Teaching & Mentoring

- **Teaching Assistant**, NJIT – Graduate: *Embedded Computing Systems*; Undergraduate: *Electronics II*. Delivered tutorials, ARM processor hardware demonstrations, and project evaluations.
- **Teaching Assistant**, IIT Bombay – *Basic Electronics Lab*, *Digital Electronics*, *Electronic Devices and Circuits*. Supervised experiments, conducted tutorials, assisted evaluations.

Languages

English	Proficient – professional and academic
German	Basic (A2) – conversational practice with native speaker
Malayalam	Native
Hindi	Conversational

Full publication list: [Google Scholar](#) | Personal page: nandasasidharan.github.io | References available on request.
Kollbrunn, June 21, 2026